PRASHANT KUMAR JHA

SENIOR JAVA BACKEND ENGINEER | TECHNICAL LEAD | MICROSERVICES | AI AGENTS / MCP

Bengaluru, India (Open to Relocate) | +91 7749994892 | jprashant2026@gmail.com

LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Hands-on **Senior Java Backend Engineer and Technical Lead with 8 years** building Spring Boot microservices, REST APIs and distributed platforms across retail, mobility, banking and enterprise CI/CD. Production experience with **Java 17, Kafka, Redis, Elasticsearch and Kubernetes**. Led technical delivery for 8 engineers at Lowe's and 10 at IBM, owning architecture, development and reviews. Optimized Elasticsearch-backed API performance using Redis caching and developed an AI-assisted vulnerability remediation platform that reduced quarterly team capacity allocated to security remediation from **20% to 5%**.

TECHNICAL SKILLS

Languages & Frontend: Java 17 (production), Java 8, React (internal tools and prototypes)

Backend & Architecture: Spring Boot, Spring, REST APIs, Microservices, Distributed Systems, System Design, API Design, Event-Driven Architecture, Hibernate, Concurrency, Spring Cache

Data & Messaging: Kafka, Redis, MQTT, PostgreSQL, Elasticsearch, MongoDB, MySQL, Oracle

Cloud, Delivery & Tools: Kubernetes, Docker, AWS S3, CI/CD, Argo CD, Argo Workflows, Maven, Git, SonarQube, Grafana, Kibana, Prometheus

Automation, Security & AI: Snyk, AI agents, MCP, Devin CLI, RAG-guided workflows, tool/function calling, Copilot, Handlebars, Pigeon, k6/PRFX, JUnit 5, JaCoCo, BDD; LLM APIs (POC)

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Lowe's India | Bengaluru | Jan 2025 – Present

Estate Management Platform | Java 17, Spring Boot, Kafka, Redis, PostgreSQL, Elasticsearch, MQTT, Kubernetes

- **Led hands-on technical delivery for 8 engineers**, driving backend design, implementation, code reviews and mentoring; delivered **30+ REST APIs and microservices** for enterprise retail platforms.
- Architected fault-tolerant, PostgreSQL-backed store-edge services supporting localized operations across **1,621 retail stores**; developed event-driven integrations using Kafka, MQTT, Redis and Elasticsearch.
- Redesigned Elasticsearch-backed read APIs with a **shared Redis cache-aside layer** across Kubernetes pods; implemented per-store incremental refresh, TTL, daily refresh, full/selective POST refresh and Redis/Elasticsearch fallback paths.
- Reduced measured **p99 API latency from 2.5–3 seconds to 170–180 ms (~93–94%)**; authored k6 scripts (5 VUs/API, 3 minutes), ran tests through PRFX and verified per-API performance in Grafana.
- Load-tested Nimbus staging APIs with k6 at 50 peak virtual users over 5 minutes (27,286 requests, ~91 req/s), recording 95 ms overall p99 latency and zero failed requests or breached thresholds.
- **Designed and built an internal Snyk vulnerability platform from scratch** for Java, Python and npm repositories, integrating Snyk APIs, Tachyon/Estate repository views, Confluence MCP archival dates, local fix validation and HBS/Pigeon reporting.
- Developed Maven **parent, standalone and manual remediation workflows** with AI reasoning, RAG-based standards, dependency-tree/compile validation and Devin CLI; retained developer-controlled Git changes and senior code review.
- Reduced quarterly engineering capacity allocated to vulnerability remediation **from 20% to 5% (75% relative reduction)**, replacing repeated manual version trials and verification with guided workflows.
- Guided controller-level BDD test restructuring and Elasticsearch connector coverage, increasing JaCoCo-measured coverage **from 46% to 80%**.
- Developed **Grafana, Kibana and Prometheus** monitoring dashboards for backend observability.
- Built secure **device-login and secret-creation workflows** adopted as an org-wide standard.
- Reduced deployment duration **from 6–7 minutes to 50–60 seconds** through CI/CD automation.
- Supported production incident response during BFCM; the retail platform processed **2.3 million transactions across the full event**.

Senior Software Engineer | IBM India | Jul 2023 – Dec 2024

Enterprise CI/CD Platform (LOCI) & Tachyon Nimbus | Java 17, Spring Boot, Kubernetes, Kafka, Argo CD, Argo Workflows

- **Led technical delivery for 10 engineers** on a Java 17/Spring Boot Kubernetes-based enterprise CI/CD control plane, contributing to architecture, hands-on development, code reviews and mentoring.
- Designed Kafka-backed deployment orchestration with Argo Workflows and introduced Argo CD automation, reducing reliance on third-party delivery tooling such as Jenkins.
- Resolved complex pre-production and production issues across distributed deployment environments and supported service reliability.
- Built 10+ Spring Boot REST APIs and Kafka-based event-driven integrations on the Estate Management Platform (Tachyon Nimbus) on Kubernetes, while continuing to support LOCI delivery.
- Recognized with the IBM Best Performer Award (Jun 2024) for consistent, high-quality delivery while deployed on-site at Lowe's, leading to a full-time offer and conversion in Jan 2025.

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior Software Engineer | HCLTech | Oct 2021 – Jun 2023

Parking Assistance, Mobility Marketplace and DeepSite | Java 8, Spring Boot, Hibernate, MySQL, Redis, AWS S3

- Developed **15+ REST APIs** and backend service flows for a Parking Assistance platform supporting approximately **7,000–8,000 concurrent users** at peak load.
- Implemented Spring Cache and Redis for backend services, reducing the share of slow responses **from 39% to 12%** during peak hours.
- Built Java/Spring Boot services for Mobility Marketplace integrations involving business APIs, mobility hardware and AWS S3 asset storage; supported production releases.

Software Engineer / Associate Software Engineer | CGI | Nov 2018 – Oct 2021

CGI Trade 360 | Java 7, Spring Boot, Hibernate, Oracle 12c, REST Web Services

- Developed and enhanced CGI Trade360 **international banking transaction-processing workflows** trade business rules and Hibernate/Oracle-backed REST services; resolved production and UAT defects.

EDUCATION

B.Tech, Computer Science Engineering | College of Engineering Bhubaneswar (BPUT)

CERTIFICATIONS & RECOGNITION

Certifications: Career Essentials in Generative AI – Microsoft & LinkedIn | SAFe 5 Certification – Scaled Agile | HPE Certified Java Developer
Awards: Raise the Roof Award – Lowe's India (Jan 2026, Mar 2026) | Team Excellence Award – Lowe's India (Feb 2026) | Best Performer Award – IBM (Jun 2024) | Key Contributor Award – CGI (2020, 2021)